INCH-POUND

MIL-DTL-3903G
1 June 2020
SUPERSEDING
MIL-DTL-3903F
17 May 2013

DETAIL SPECIFICATION

WIRE ROPE ASSEMBLIES, SINGLE LEG, (SLING TYPE)

Reactivated after 27 October 2009 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers wire rope assemblies, single leg with plain end loops.

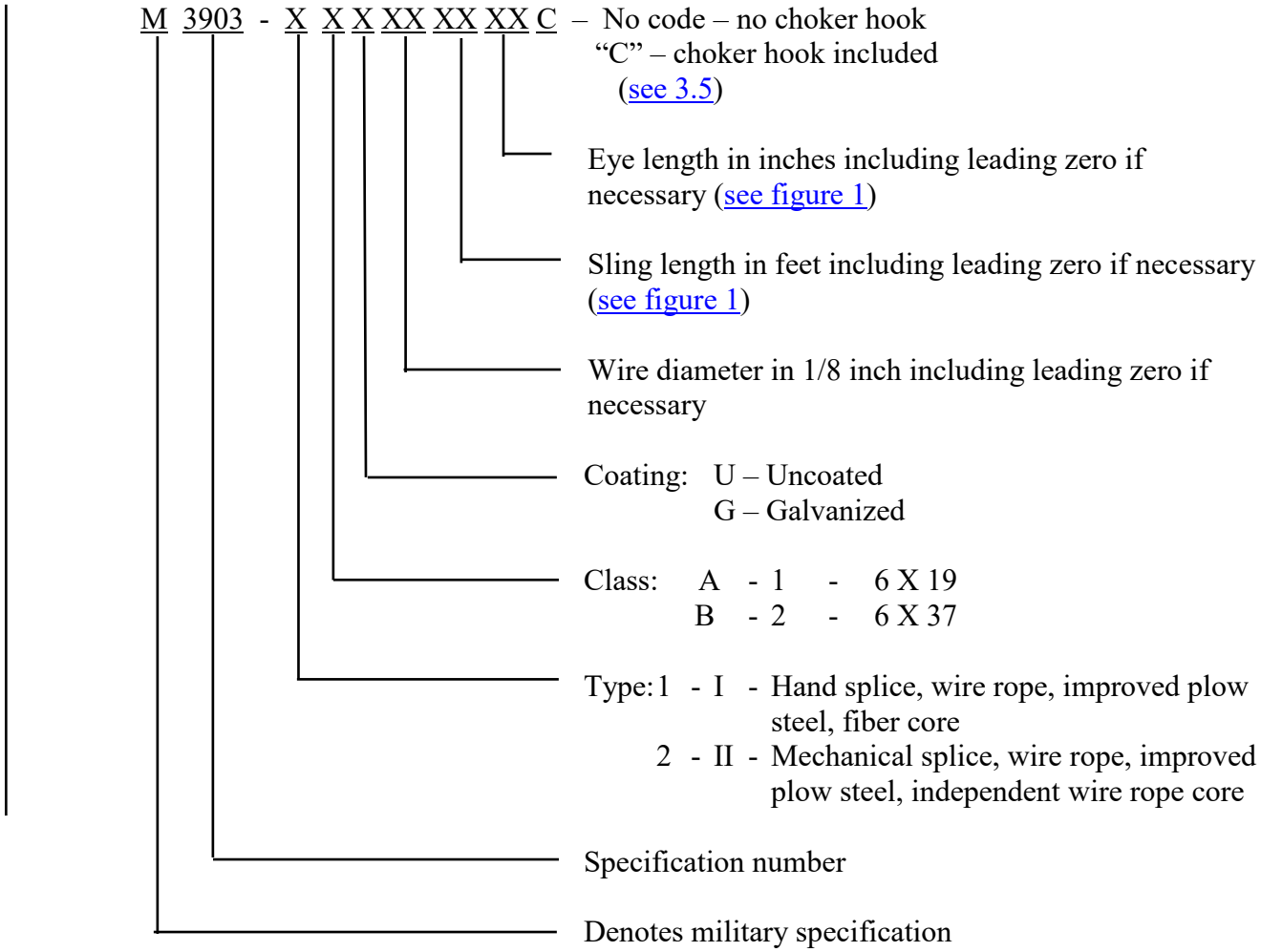
1.2 Classification. The wire rope assemblies are of the following types and classes, as specified (see 6.2):

Type I	-	Hand splice, wire rope, improved plow steel, fiber core
Type II	-	Mechanical splice, wire rope, improved plow steel, independent wire rope core
Class 1	-	6 by 19 (RR-W-410, Type I, Class 2, assemblies with wire rope diameter of 1-1/8 inch and under)
Class 2	-	6 by 37 (RR-W-410, Type I, Class 3, assemblies with wire rope diameter of 1-1/4 inch and over)

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency Aviation - VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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1.3 Part or identifying number (PIN). The PIN to be used for wire rope assemblies acquired to this specification is created as follows:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

RR-W-410	- Wire Rope and Strand

COMMERCIAL ITEM DESCRIPTION

A-A-52540	- Fittings, Wire Rope

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-DTL-13220	- Hook, Sliding Choker (for use with Wire Rope)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-889	- Dissimilar Metals

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of this document are available online at https://www.astm.org/.)

SAE INTERNATIONAL

SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)

(Copies of this document is available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2 and 6.3.

3.2 Description. Wire rope assemblies (hereinafter called "slings") shall be as shown in figure 1 and specified herein.

3.2.1 Characteristics. Each sling shall be fabricated from one continuous piece of new and unused wire rope with a loop at each end, secured with either a hand or swage splice, as shown in figure 1.

3.3 Material. Materials shall be as specified herein. Materials not definitely specified shall be selected by the contractor and shall be new and unused and subject to all provisions of this document.

3.3.1 Material deterioration and control. The slings shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Wire rope. Wire rope shall be of improved plow steel and conform to RR-W-410, type I, class 2 or class 3 as applicable, for preformed, regular lay, and shall be the size and finish specified in table I.

TABLE I. Wire rope sizes and finish for types I and II slings.

Item	Wire rope diameter	Finish
Types I and II		
Class 1	3/8 through 1-1/8 inches	Uncoated or galvanized as specified (see 3.7)
Class 2	1-1/4 through 2 inches	Uncoated or galvanized as specified (see 3.7)

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3.4 Design.

3.4.1 Type I. Type I slings shall conform to table II and shall consist of one continuous length of steel wire rope as specified in 3.3.2 with hand spliced eye loops on each end as shown in figure 1.

TABLE II. Strength requirements in tons (2,000 pounds) for type I, classes 1 and 2 slings.

Diameter	RC[1] (one leg vertical pull)		Proof load		Minimum breaking strength[2]		Splice efficiency
Inches	Tons, uncoated	Tons, galvanized	Tons, uncoated	Tons, galvanized	Tons, uncoated	Tons, galvanized	Percent
Class 1:							
3/8	1.1	0.99	2.2	1.98	4.96	4.47	88
1/2	1.8	1.60	3.6	3.20	8.51	7.66	86
5/8	2.8	2.50	5.6	5.00	12.97	11.65	84
3/4	3.9	3.50	7.8	7.00	18.05	16.23	82
7/8	5.1	4.60	10.2	9.20	23.83	21.45	80
1	6.7	6.00	13.4	12.00	30.93	27.84	80
1-1/8	8.4	7.60	16.8	15.20	38.92	35.03	80
Class 2:							
1-1/4	9.8	8.80	19.6	17.60	45.51	40.96	80
1-3/8	12.0	11.00	24.0	22.00	54.83	49.36	80
1-1/2	14.0	13.00	28.0	26.00	65.05	58.54	80
1-5/8	16.0	14.00	32.0	28.00	76.20	68.58	80
1-3/4	19.0	17.00	38.0	34.00	88.06	79.18	80
1-7/8	22.0	20.00	44.0	40.00	100.65	90.58	80
2	25.0	22.00	50.0	44.00	113.95	102.86	80

[1] RC – Rated capacity is the maximum recommended load based on a safety factor of 5.
[2] Strength requirements computed on the efficiency of the eye splice and the 7-1/2 percent for testing variation.

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3.4.1.1 Seizing. Seizing shall be of strand conforming to RR-W-410, type V, class 1, or wire and shall be iron or annealed steel, as specified (see 6.2). Construction and dimensions shall be as specified in table III. A seizing of strand or wire shall cover the splice and adjacent strands, which were unlaid during splicing. Seizing shall be tightly wound, each wrap of seizing laying against the preceding wrap, the ends twisted together, and laid against the rope.

TABLE III. Seizing strands size and construction.

Wire rope diameter (Inch)	Seizing strand		Seizing wire minimum diameter (Inch)
	Diameter (Inch)	Construction	
3/8	1/16	1 by 7	0.080
1/2	3/32	1 by 7	0.080
5/8	1/8	1 by 7	0.080
3/4	1/8	1 by 7	0.104
7/8 and larger	5/32	1 by 7	0.104 and larger

3.4.1.1.1 Length of serve. The length of serve shall be 16 times the wire rope diameter (see figure 1).

3.4.1.2 Hand tucks. Hand-tucked splices shall be made by forming a loop and splicing the dead end into the live end of the rope. Each dead-end strand shall be given one forming tuck and three full tucks around the same strand in the body of the rope. One additional full tuck shall be made when splicing flexible wire ropes such as the 6 by 37 classification. Splices shall be tightly tapped to remove high spots and allow the strands to mesh.

3.4.1.3 Forming tucks. Forming tucks shall be made by prying two adjacent strands apart, inserting a dead-end strand into the opening, and passing the strand under one, two, or three adjacent strands in the body of the rope. The dead-end strand shall be set or locked tightly.

3.4.1.4 Full tucks. Full tucks shall be made by inserting a dead-end strand under a strand in the body of the wire rope and rotating it 360 degrees. The tucked strand shall be set or locked tightly. Each subsequent full turn of the dead-end strand around the live-end strand constitutes an additional full tuck.

3.4.1.5 Setting or locking. Setting or locking of a dead-end strand shall be accomplished by pulling the strand end in under considerable force. At the same time, a marlin spike, inserted in the same opening in the body of the rope, shall be rotated about the axis of the rope back to the start of the splice or to the previous tuck.

3.4.1.6 Tensile strength. Minimum tensile strength shall be as specified in table II.

3.4.2 Type II. Type II slings shall consist of one continuous length of steel wire rope as specified in 3.3.2, table IV, and with mechanically spliced eye loops on each end as shown in figure 1.

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TABLE IV. Strength requirements in tons (2,000 pounds) for type II, classes 1 and 2 slings.

Diameter	RC[1] (one leg vertical pull)		Proof load		Minimum breaking strength[2]		Splice efficiency
Inches	Tons, uncoated	Tons, galvanized	Tons, uncoated	Tons, galvanized	Tons, uncoated	Tons, galvanized	Percent
Class 1:							
3/8	1.2	1.1	2.4	2.2	5.90	5.31	95.0
1/2	2.2	2.0	4.4	4.0	10.35	9.31	95.0
5/8	3.4	3.1	6.8	6.2	16.11	14.50	95.0
3/4	4.9	4.4	9.8	8.8	23.04	20.74	95.0
7/8	6.6	5.9	13.2	11.8	31.14	28.03	95.0
1	8.5	7.6	17.0	15.2	40.41	36.37	95.0
1-1/8	10.0	9.0	20.0	18.0	49.43	44.68	92.5
Class 2:							
1-1/4	12.0	11.0	24.0	22.0	57.84	52.05	92.5
1-3/8	15.0	13.0	30.0	26.0	69.74	62.76	92.5
1-1/2	17.0	16.0	34.0	32.0	82.69	74.42	92.5
1-5/8	20.0	18.0	40.0	36.0	97.12	87.41	92.5
1-3/4	24.0	21.0	48.0	42.0	112.00	100.80	92.5
1-7/8	27.0	24.0	54.0	48.0	127.75	114.97	92.5
2	30.0	27.0	60.0	54.0	144.37	129.94	92.5

[1] RC – Rated capacity is the maximum recommended load based on a safety factor of 5.
[2] Strength requirements computed on the efficiency of the eye splice and the 5 percent for testing variation.

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3.4.2.1 Mechanical splices. A mechanical splice shall be fabricated by pressing or swaging one or more metal sleeves over the rope junction.

3.4.2.2 Fittings. Swage sleeves shall conform to A-A-52540 or other swage sleeves that conform to the quality assurance provisions of A-A-52540.

3.4.2.3 Strand ends. Spliced strand ends shall be neatly cut. Burning shall not be permitted.

3.4.2.4 Tensile strength. Minimum tensile strength shall be as specified in table IV.

3.5 Choker hooks. When specified (see 6.2), a sliding choker hook shall be fitted between the loops of each sling. Hooks shall be the applicable size for the wire rope and shall conform to MIL-DTL-13220.

3.6 Dimensions. The length of the slings shall be the length from bearing point to bearing point when the loops lay the natural and the length of the loop is approximately twice the width of the loop (see figure 1). The wire rope diameter and the length of the sling and loops shall be as specified in the contract (see 6.2). The slings shall not vary from the length specified more than ± 2 diameters of the wire rope used, or ± 0.5 percent of sling lengths, whichever is greater.

3.7 Finish. Types I and II slings shall be finished as specified (see 6.2). When zinc-coated rope is specified, the metal sleeves of type II slings shall be zinc-coated in accordance to ASTM B633, Fe/Zn 25, type II finish or cadmium plated in accordance with SAE AMS-QQ-P-416, class 1, type II.

3.8 Identification. Identification shall be in accordance with MIL-STD-130. A metal tag shall be securely attached to each sling. Metal tags shall be corrosion resistant, 0.016 inch minimum thick steel, securely fastened adjacent to one of the eye spliced loops by corrosion resistant 0.047 inch minimum diameter wire. The tag shall have the following information etched or punched thereon:

WIRE ROPE ASSEMBLY, SINGLE LEG (SLING TYPE)

NSN_____ Rope dia._____ Grade_____
Length of assembly_____
Loop size_____
Contract no._____ Date_____
Manufacturer_____
Rated Capacity in short tons (2,000 pounds) _____

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3.9 Workmanship. The slings shall conform to the quality and grade of product as specified in RR-W-410, and shall be free from imperfections such as protruding strands, sharp edges, burns, burrs, and corrosion.

3.10 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 Examination. Two first article slings shall be examined as specified in 4.4.1. Presence of one or more defects shall be cause for rejection of both slings.

4.2.2 Tests. Two first article slings shall be tested as specified in 4.4.2, with one sling undergoing the proof test as specified in 4.4.2.1, and the other sling tested to destruction as specified in 4.4.2.3. Failure of either test shall be cause for rejection.

4.3 Conformance inspection.

4.3.1 Lot. A lot shall consist of all slings of the same type, class, finish, and wire rope diameter at one time for acceptance.

4.3.2 Sampling. Unless otherwise noted in the contract (see 6.2) sampling for examination shall be in accordance with an “accept at zero” (c=0) sampling plan. See 6.3 for additional information. For destructive testing, three samples from each lot of 100 or fewer slings produced shall be subjected to the test specified 4.4.2.3.

4.3.3 Examination. Samples selected in accordance with 4.3.2 shall be examined for defects specified in 4.4.1.

4.3.4 Tests. Samples selected in accordance with 4.3.2 shall be tested in accordance with 4.4.2. Failure of either test shall be cause for rejection of the entire lot represented by the sample.

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4.4 Inspection procedure.

4.4.1 Examination. Slings shall be examined as specified herein for the following defects:

Major

101. Material not as specified or the materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.
102. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
103. Finish plan not adequate or the contractor does not have documentation available for identification of material, material finishes, or treatments.
104. One or more wires broken or severed.
105. Wires not completely tucked into splices or swages.
106. Hand splices not as specified.
107. Mechanical splices not as specified.
108. Dimensions not as specified.
109. Workmanship not as specified.

Minor

201. Protruding wire strand ends.
202. Seizing not as specified.
203. Strand ends cut by burning.
204. Identification marking illegible, incomplete, or missing.

4.4.2 Tests.

4.4.2.1 Method of testing. Testing shall be done prior to any seizing when this is specified. The proof and destructive tests shall be a single-leg straight pull method. The slings shall be placed on a tensile testing machine using mandrels having a minimum diameter of four times the diameter of the wire rope used in a uniform rate. Load shall be applied through the movable mandrel at a uniform rate. An alternate test method may be used for heavy duty slings, using a hoist or crane. One terminal of the sling shall be attached to the hook and the other to the test load.

4.4.2.2 Proof load. Slings shall be tested for proof as specified in 4.4.2.1. Slippage of the swage sittings or nonconformance to the applicable minimum proof load specified in tables II or IV shall constitute failure of this test.

4.4.2.3 Ultimate tensile strength. Slings shall be tested as specified in 4.4.2.1 for ultimate tensile strength. Nonconformance to the applicable minimum breaking strength specified in tables II or IV shall constitute failure of this test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The slings covered by this specification are intended for use in cargo and material handling operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class (see 1.2).
c. When first article is required for inspection and approval and the number of units required (see 3.1).
d. Whether seizing should be strand or wire, and of iron or annealed steel (see 3.4.1.1).
e. When slings should be fitted with choker hooks (see 3.5).
f. Wire rope diameter (see 3.6).
g. Length of sling (see 3.6).
h. Length of loops (see 3.6).
i. Wire rope finish required (see 3.7).
j. Inspection level if different than specified (see 4.3.2).
k. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first produced sling. The first article should consist of two or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Sling number to PIN cross-reference. The previous PIN only designated type and class of the sling. Actual sling dimensions, wire rope coating or choker hook installation were not included. Below is a cross-reference from the previous dimensions by sling number to the new PIN. Note that all listed sizes are class 1 per table I and that the type (X) and the coating (Y) are not identified.

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TABLE V. Sling number to PIN cross-reference.

Sling number Per MIL-DTL-3903F	PIN per 1.3
1	M3903-X-A-Y-04-03-04
2	M3903-X-A-Y-04-07-08
3	M3903-X-A-Y-04-20-10
4	M3903-X-A-Y-04-30-10
5	M3903-X-A-Y-04-40-10
6	M3903-X-A-Y-04-50-10
7	M3903-X-A-Y-05-30-12
8	M3903-X-A-Y-05-40-12
9	M3903-X-A-Y-05-50-12
10	M3903-X-A-Y-06-07-12
11	M3903-X-A-Y-06-20-12
12	M3903-X-A-Y-06-30-12
13	M3903-X-A-Y-06-40-12
14	M3903-X-A-Y-06-50-12
15	M3903-X-A-Y-07-20-13
16	M3903-X-A-Y-07-30-13
17	M3903-X-A-Y-07-40-13
18	M3903-X-A-Y-07-50-13
19	M3903-X-A-Y-08-20-14
20	M3903-X-A-Y-08-30-14
21	M3903-X-A-Y-08-40-14
22	M3903-X-A-Y-08-50-14
23	M3903-X-A-Y-09-20-18
24	M3903-X-A-Y-09-50-18
25	M3903-X-A-Y-03-12-08C
26	M3903-X-A-Y-03-16-10C
27	M3903-X-A-Y-04-12-10C
28	M3903-X-A-Y-05-20-12C

6.5 Accept at zero sampling plans. Acceptable “accept at zero” (c=0) sampling plans include but are not limited to MIL-STD-1916 or ASQ H1331, table 1 unless otherwise specified by the contact. If the applicable drawing, specification, standard, or quality assurance provision (QAP) specifies Critical, Major and/or Minor attributes, it is recommended that they be assigned Verification Levels of VII, IV and II for MIL-STD-1916, AQLs of 0.1, 1.0 and 4.0 for ASQ H1331, or similar inspection levels for other plans respectively. Unspecified attributes should be considered as Major. For MIL-STD-1916 the manufacturer may use the attribute or variable inspection method at their option. MIL-STD-105/ASQ Z1.4 may be used to set sample lot size but acceptance is at zero non-conformances.

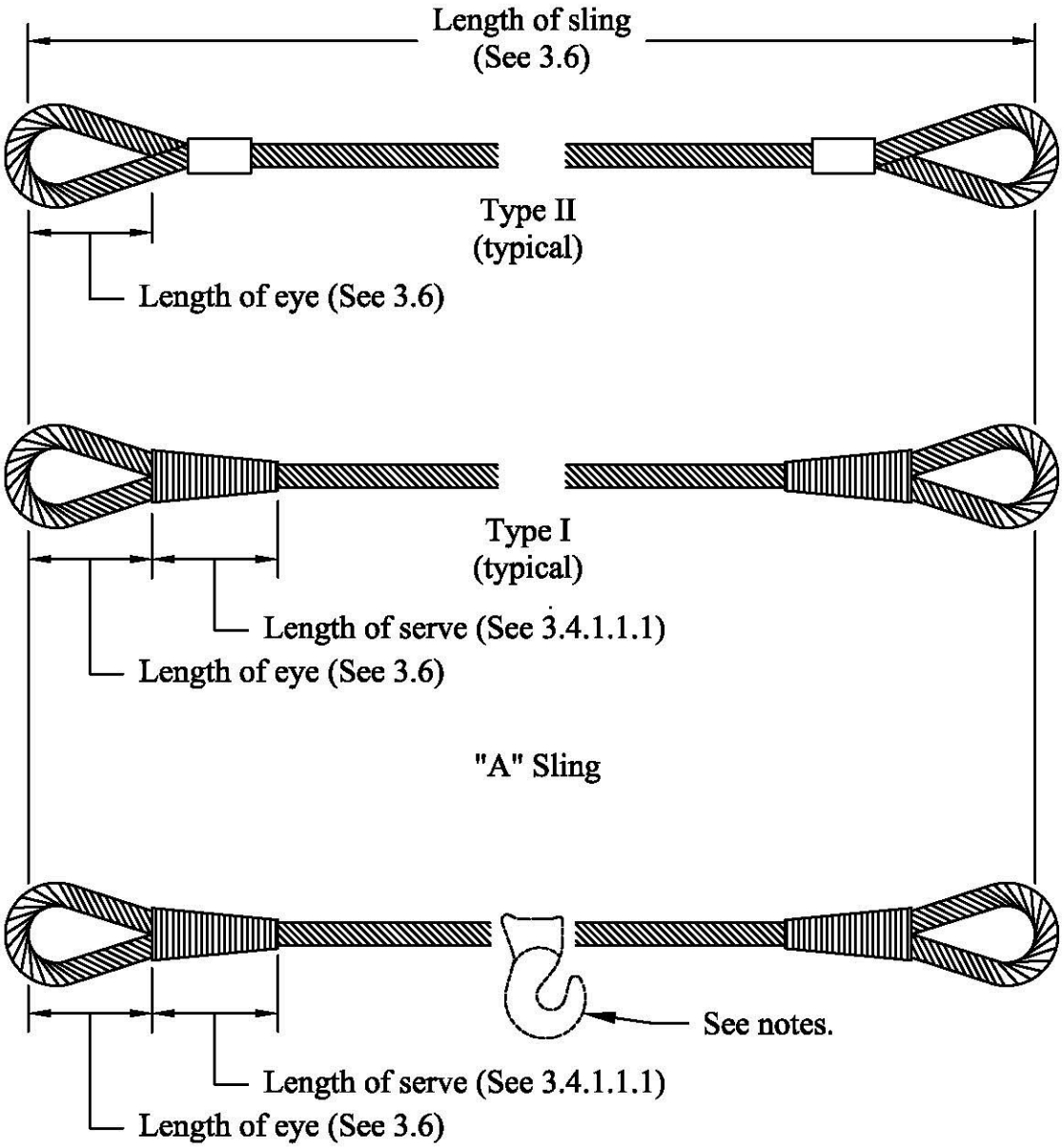
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6.6 Subject term (key word) listing.

Cadmium
Hand splice
Mechanical splice
Swage sleeve
Zinc

6.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notions. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. "B" sling is furnished with a choker hook.
2. Hook size shall match up with rope size.
3. Sling may be type I or II as specified

FIGURE 1. Mechanical and hand splices.

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Preparing Activity:
DLA - GS5

(Project 4010-2020-001)

Review Activities:
Army - MI
Navy - OS, SH
Air Force - 19

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